Embedded Systems International - Project Proposal
Autonomous CyRide

Problem Statement

CyRide is the public transportation system utilized by the city of Ames, IA and Iowa State University, providing residents and students with over 13 different bus routes. According to the CyRide website, their busiest route (Orange 23) carried over 1.7 million passengers in 2019. But this increase in demand for CyRide requires a supply of drivers to manage the routes, and CyRide needs to catch up. Due to their shortage of drivers, CyRide continues to push hiring and even increase driver wages to attract personnel. And with the current job market, hiring and retaining drivers for a long period takes work. To solve this problem, we will implement a *self-driving* bus that, like the 23 Orange bus, picks up students at specific locations and drops them off at their requested stops along the route *without requiring a driver*.

For this project, we will assume the CyBot is our bus and incorporate all major aspects of our self-driving bus to make it a safe and viable option for CyRide to consider. In this sense, we will have the CyBot scan at its starting location, the Park and Ride Terminal, to determine the number of passengers boarding from the terminal. The bus then begins driving autonomously and listens for passengers to request a stop by pulling the lever. When a passenger requests a stop, the bus will stop at the upcoming bus stop. The proposed demonstration system will have four stops on this route; the Park and Ride Terminal, where passengers are initially picked up, and three other locations along the route.

While driving on the route, the bus may encounter short objects or cliffs such as rocks, debris in the road, or potholes, which it must automatically detect and avoid with appropriate sensors. In addition, the bot will continuously scan for tall objects, such as pedestrians and cars crossing the roadway. If one is detected, the bus will break and wait for the object to cross, then continue driving again all autonomously. Another major aspect of this project will be our bus's interaction with the Control Center: a computerized monitoring station for CyRide operations personnel to monitor. This system allows managers to observe information such as passenger count and incidents occurring on the roadway, such as short or tall objects or cliffs detected, as well as when passengers request a stop and when the bus arrives at a stop.

Application Narrative

Autonomous CyRide involves two different user perspectives: passengers who want to utilize the bus system without issue and management who want to observe telemetry and operations information at all times to ensure no problematic incidents occur.

For passengers, the most important thing is to be dropped off at the proper location. For instance, a student is picked up at the starting point and wants to go to BUS STOP 2. This student wants the bus to properly identify them and where they are going. The student wants the bus to stop at the appropriate bus stop. If the student goes to a class or an appointment, they expect the bus to be on time, so the bus should only stop when asked, and if someone misses their stop, they expect the bus to have an option of stopping at the next bus stop instead of being forced to complete a round trip.

For the system manager in the control center, quickly receiving accurate information is crucial. A system manager will want the bus to send accurate information about the number of passengers and their requests. They will also expect it to send information about traffic, time, and accidents.

Empathy Map

This diagram describes the perspectives of the passengers themselves

Do: ● Take public transportation to get where they need to go. ● Live in an area where public transportation is needed. ● Take the bus because it is more affordable	**Think:** ● "I wish this bus driver would skip locations that people are not getting off at." ● "I need a reliable way to get to my destinations." ● "I need public transport in my area because of my living conditions in a densely populated area." ● "I can't afford to have my own car so I need public transportation."
Say: ● "I need an affordable way to get to work" ● "I can't have my own car, so I need the use of public transportation" ● "The places I need to get to require special parking permits, so I need public transport" ● "I am not a very good driver in urban areas"	**Feel:** ● Concerned if their current living conditions do not permit a vehicle of their own in the area ● Worried of their current financial situation so public transportation is a more feasible option ● Anxious in certain driving conditions so public transport feels like a safer option ● Unsure if an automated system is reliable for transportation of a large amount of people

Point of View Statements/ User Types

This User of this service/product would say:

- "There is a very long wait between buses if I happen to miss the one I need to catch, having more circulating would be nice."
- "Walking across campus to get to different classes takes too long."
- "I have a car, but there isn't enough parking available on campus."
- "I can't afford to buy a car, efficient public transportation is a necessity for me."
- "Campus is so crowded with people I don't feel comfortable driving there."
- "I would like to take the bus more often, but I don't feel safe riding when it is packed full of people."
- "Be able to monitor what the bus is doing at any time"

Users Needs

List Users needs based off Research

- Higher number of buses circulating to accommodate the amount of people that take the bus regularly.
    - [1] T. K. Albertson, “CyRide cuts weekend service hours due to staff shortage and adds $1,000 signing bonus,” *The Ames Tribune*. https://www.amestrib.com/story/news/2022/08/18/cyride-staffing-shortage-leads-cut-service-1-000-signing-bonus/10356361002/ (accessed Nov. 01, 2022).

- Alternative to having a dedicated driver for each bus to lower total cost of public transport and increase the amount of active buses.
    - [2] A. Burger, “What Might Autonomous Public Transit Look Like?,” *www.vta.org*, Dec. 06, 2021. https://www.vta.org/blog/what-might-autonomous-public-transit-look (accessed Nov. 01, 2022).

- Reliable and efficient public transportation method that users can count on.
    - [3] Administration, “The Importance of Reliable Transportation,” *OATS Transit*, Oct. 18, 2021. https://www.oatstransit.org/post/the-importance-of-reliable-transportation#:~:text=Reliable%20transportation%20is%20a%20vital (accessed Nov. 02, 2022).

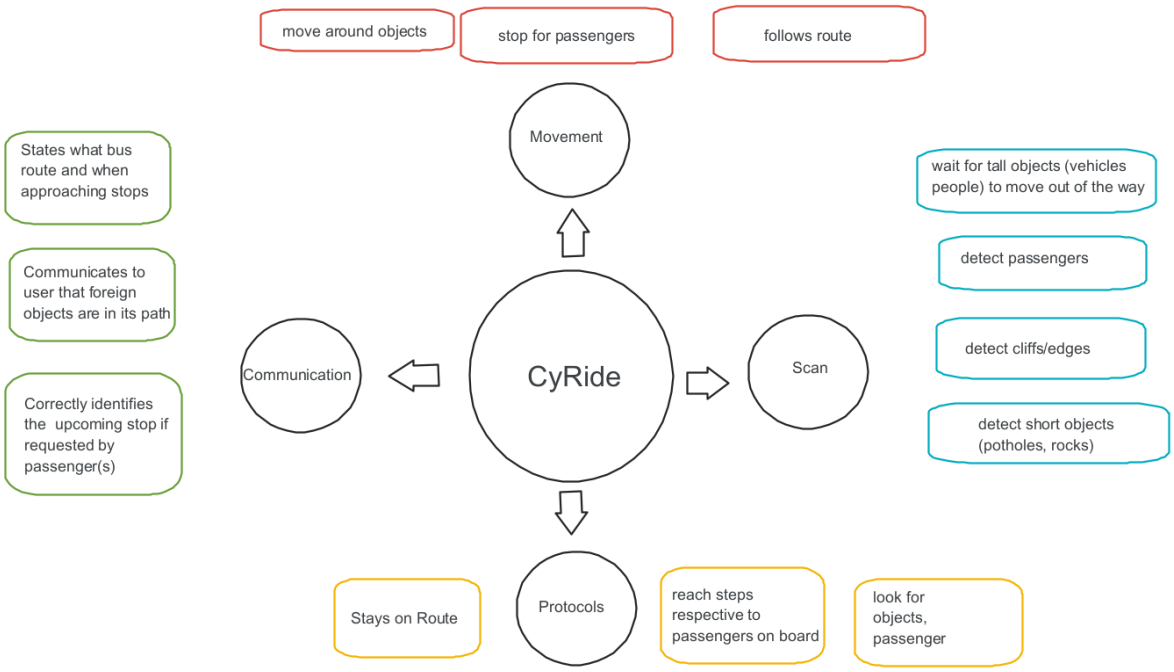
Large Functionality

Base Functionality	Mapping to Application Narrative
CyBot Movement	Automatic and manual drive passengers to their desired locations
CyBot Scan	Collect data for the control center and the CyBot auto-drive system.
CyBot Communication	Send important information to the control center and receive specific instructions
CyBot Protocols	Things we want the CyBot to do under specific conditions like a return to base in the middle of the route or avoid some objects

Mapping Functionality to CyBot Capabilities

Function	Project Equivalent	Implementation	Capabilities Used
Movement	CyBot Driving	Use the movement functions for the CyBot to get around	iRobot Open Interface
Scan	Detect passengers	We will use this function to scan for objects (passengers) at the bus stop	ADC, Interrupts, Timers, Servo, IR/PING)))
UART + WiFi + Serial Communication	Send and receive commands from the control center	Send information (Display) about the number of passengers and their desired stop location. Receive navigation instructions from the control center	UART, Interrupts, GPIO
Servo	Manipulate passenger scan area	Will be used to turn the servo for a scan of the “bus stop” and send the data to the control center.	Timers, ADC, GPIO
ADC	Raw object distance	Will be used to collect data about object distance that will then be converted into actual distance.	ADC, GPIO

Lotus Blossom Diagram



Our autonomous CyRide needs to be able to utilize various subsystems onboard to ensure the accurate amount of passengers are picked up, the bus drops off the passengers at the correct stops, and the bus keeps passengers safe during the ride, such as avoiding holes and objects (debris or pedestrians) in the road all while remaining on the route. These problems require four technical areas outlined in the diagram and large functionality above: Communication, Movement, Scanning, and Protocols. Each category outlines specific technical CyBot systems to enable each category's function in the mapping table.

Communication is responsible for alerting the control center (PC) of passenger counts and pick/up drop off at determining bus stops. The movement will entail using various sensors to move between lanes, stop passengers at the starting point, follow the predetermined bus route, and stop or move around cliffs and objects (additionally alerting the control center of problems in the road). The scan will enable automated passenger detection, finding objects on the road, and avoiding cliffs. Finally, protocols refer to the various software algorithms to determine when to execute each step or accept input from sensors to follow the correct bus procedure and repeatedly perform it until a manual override from an official is triggered.

Prototype

A. User-Centered Sketch

This sketch outlines the basic operating procedure for autonomous CyRide and how it relates to the prototype test.

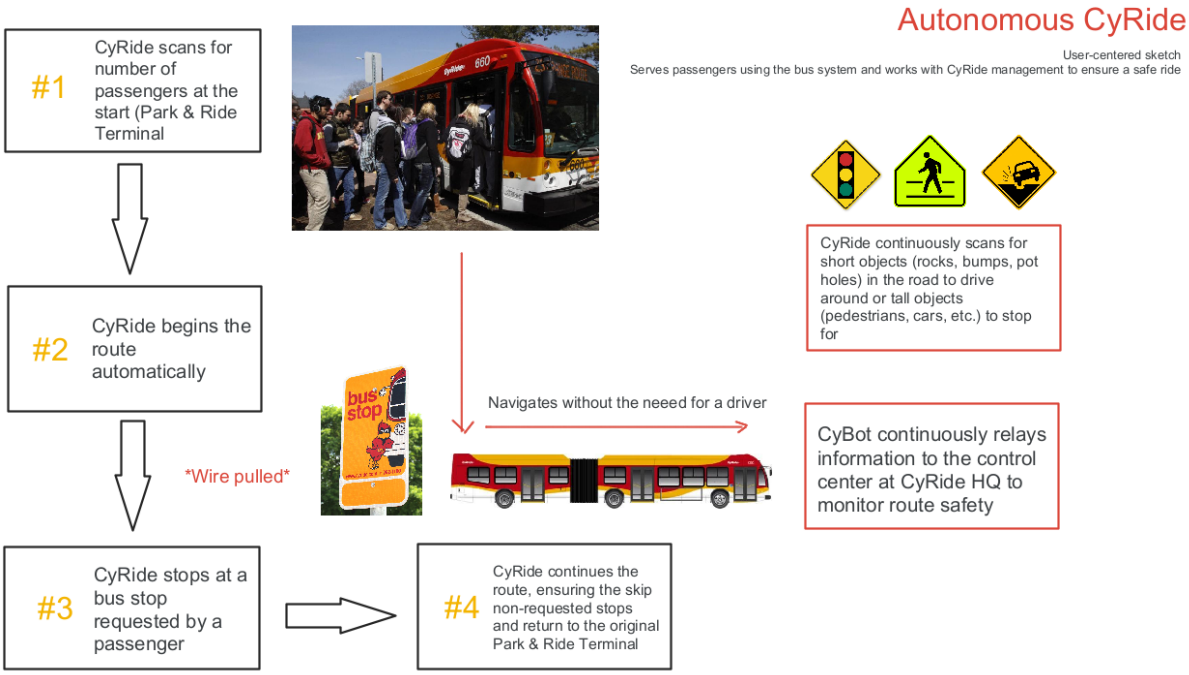
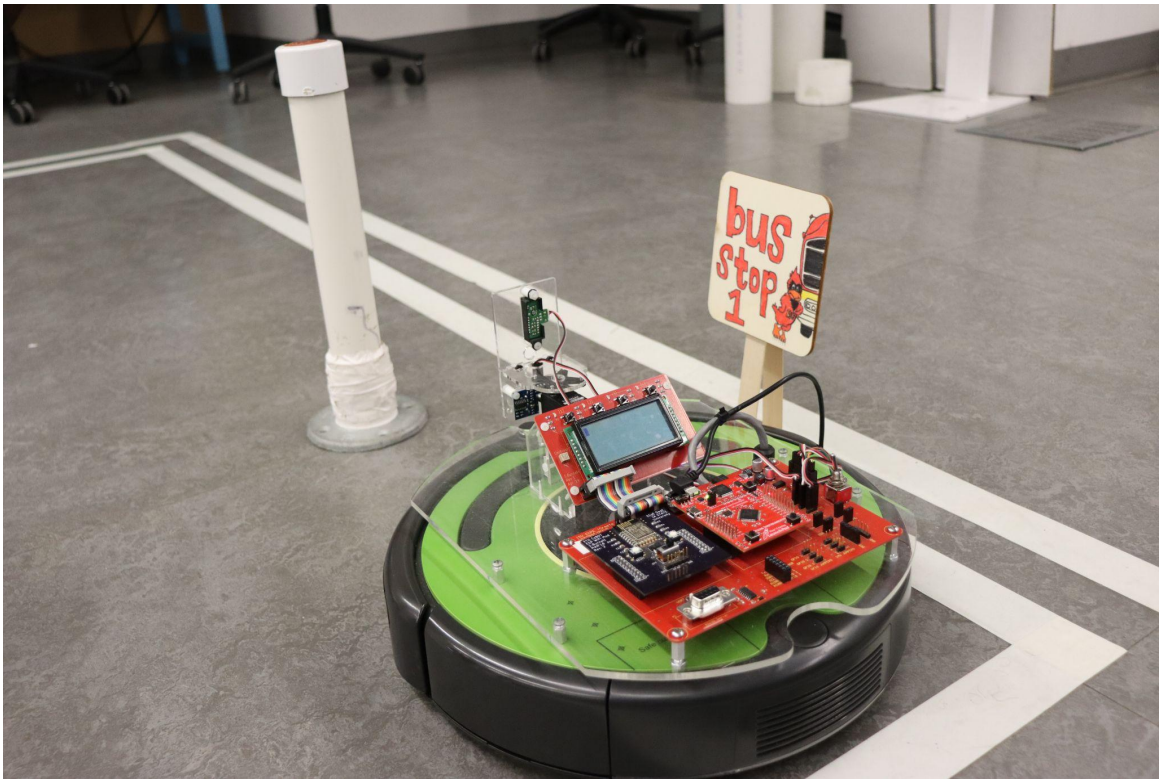


Figure A-2: The CyBot used as a prototype CyRide on the test field



B. Test-Field Sketch

This sketch outlines the details of the prototype bus route for our CyRide automation test.

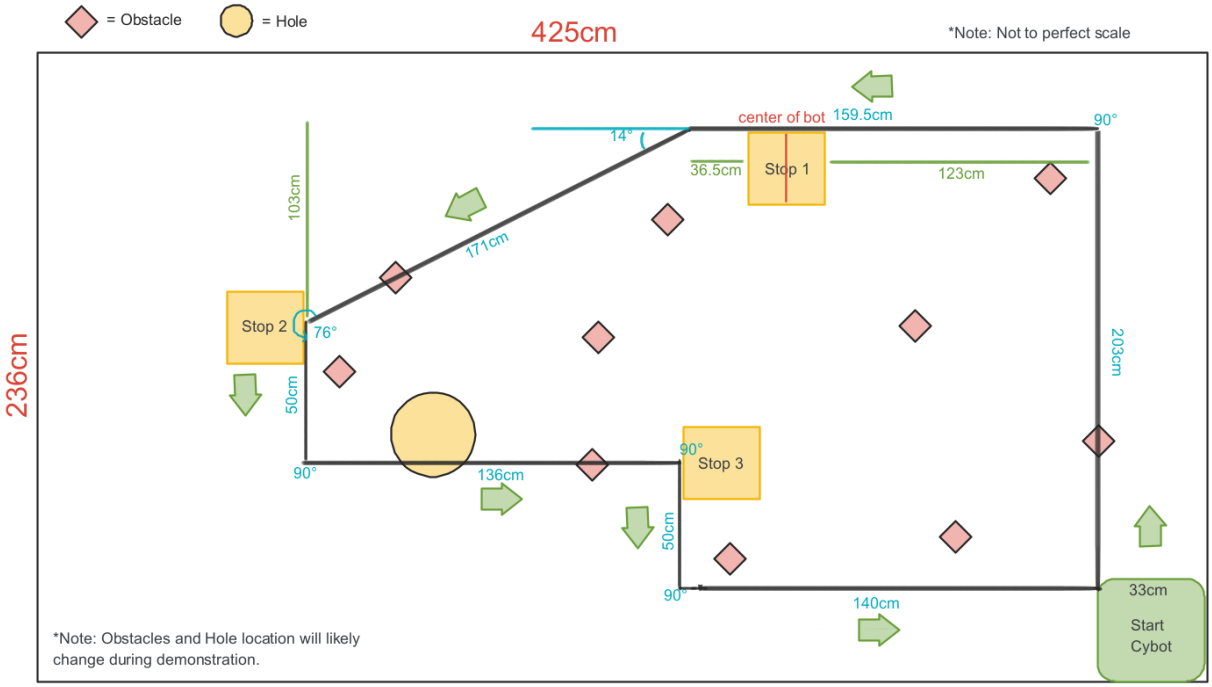
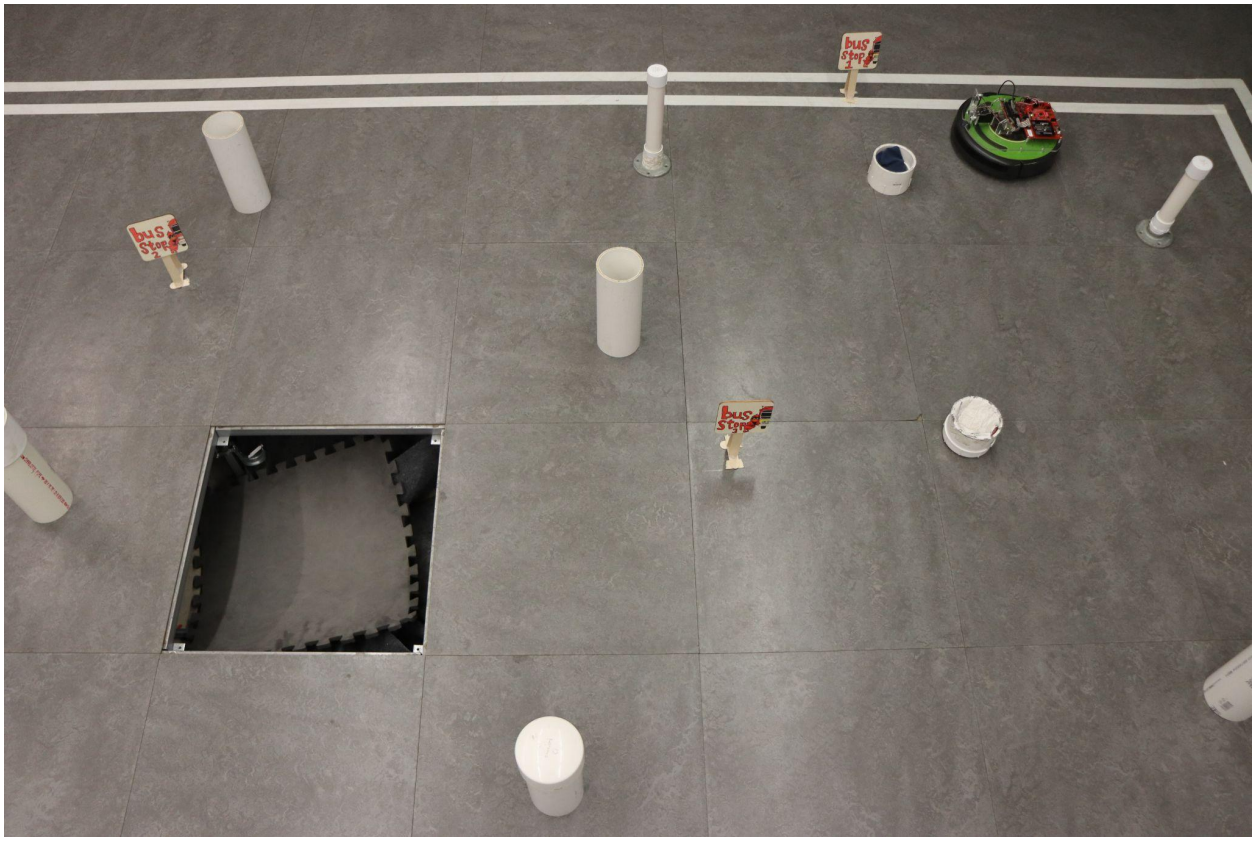
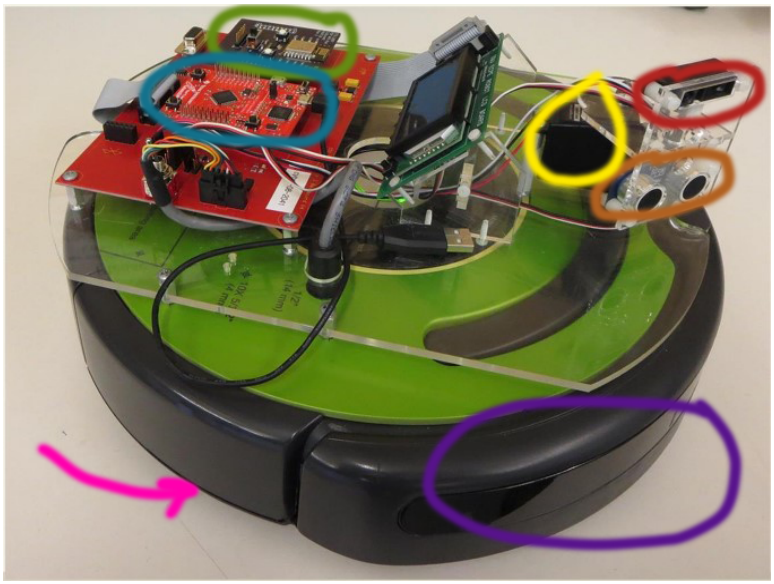


Figure B-2: The test-field set up according to the sketch



C. Technical-Centered Sketch

This sketch outlines the various components of the CyBot prototype and how they correspond to the product end goal.



TM4C123GH6PM Microcontroller - Main program CPU and memory to autonomously carry out instructions such as driving, scanning for passengers, and keeping track of stops. This is the primary computer component that allows the prototype to execute instructions.

iRobot Open Interface - Wheels located under the CyBot allow for forward/backward movement and clockwise/counterclockwise rotation

UART Serial Communication + WiFi Module - This module (microcontroller + separate chip) allows wireless communication between the CyBot and the control station that sends destination and navigational information to the CyBot based on the requested stops passengers input before boarding

Servo Motor - Allows 180 degree movement of CyBot sensors to detect passengers across an entire platform at a bus stop (for accounting each passenger) and detection of objects in the roadway

IR Sensor - Detects passengers of all sizes and shapes easily and quickly to allow the CyBot to know how many passengers to account for, Detects cars or pedestrians (tall objects) in the road

PING))) Sensor - Additional passenger detection, acts as a contingency sensor and second data point for detecting passengers, i.e. for if the IR sensor were to fail during operation or improperly detect a nonexistent passenger

Bump and Cliff Sensor - Incorporates additional safety features on the prototype, such as the ability to quickly detect obstacles, potholes, and other road impurities, to avoid collisions and report unsafe travel conditions or traffic to the control station.

Team Collaboration:

Summary Collaboration: Everyone one as a whole contributed in a well and meaningful way for achieving our end result in this project proposal. We all provided unique and interesting perspectives that helped further improve our image of what we are set out to attain. All group members worked well together and cooperatively researched to further improve and implement all aspects of the robot we have learned thus far in class. Team members' specific contributions are outlined as follows:

Team Member	Contribution
Tyler Smith	Helped create a layout for the project proposal. Talked to the team about ways to brainstorm ideas for the project and make it unique according to everyone's feedback. Researched and created an empathy map to understand the potential users' emotional motivations for needing this product. Helped create layout of the path taken by the CyRide with addition of stops. Designed aesthetic of creating small wooden signs to help identify stops. Helped with critiquing and commenting on code that progressed the overall goal.
Henry Shires	Conducted Brainstorming and revised ideas with the team. Wrote source code and managed version control (GitHub) to solve software-related problems. Managed project tasks and organized codebase to increase team efficiency
Emile Albert Kum Chi	Wrote the problem statement, application narrative, and documented aspects of the CyBot to implement in the project prototype. Measured the CyBot route and did the necessary calculations to ensure the CyBot completes its route. Assisted with code debugging and testing problems.
Cael O'Regan	Researched sources to provide insights on user needs and developed corresponding POV statements to summarize each user segment's background and perspectives. Implemented functions that allowed the CyBot to autonomously navigate around obstacles and return to the specified path.